Lab No: 06

SUPPORT VECTOR MACHINE (SVM) CLASSIFIER

Procedure:

Source Codes and Outputs:

```
from sklearn.datasets import load_breast_cancer
data = load_breast_cancer(as_frame=True)
X = data.data
y = data.target
```

```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

numeric_features = X.select_dtypes(include=["int64", "float64"]).columns
categorical_features = X.select_dtypes(include=["object", "category"]).columns

numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])
```

[2] ✓ 0s

```
categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_transformer, numeric_features),
    ("cat", categorical_transformer, categorical_features)
])
```

[3] ✓ 0s

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
X_train = preprocessor.fit_transform(X_train)
X_test = preprocessor.transform(X_test)
```

[4] ✓ 0s

```
from sklearn.model_selection import GridSearchCV
```

[4] ✓ 0s

```
from sklearn.model_selection import GridSearchCV
from sklearn.svm import SVC
param_grid = {
    "C": [0.1, 1, 10],
    "kernel": ["linear", "rbf"],
    "gamma": ["scale", "auto"]
}
grid = GridSearchCV(SVC(), param_grid, cv=5, scoring="f1")
grid.fit(X_train, y_train)
```

▸ **GridSearchCV** ⓘ ?

▸ **best_estimator_: SVC**

▸ SVC ?

Variables Terminal

[5] ✓ 0s

```
y_pred = grid.predict(X_test)
```

[6] ✓ 0s

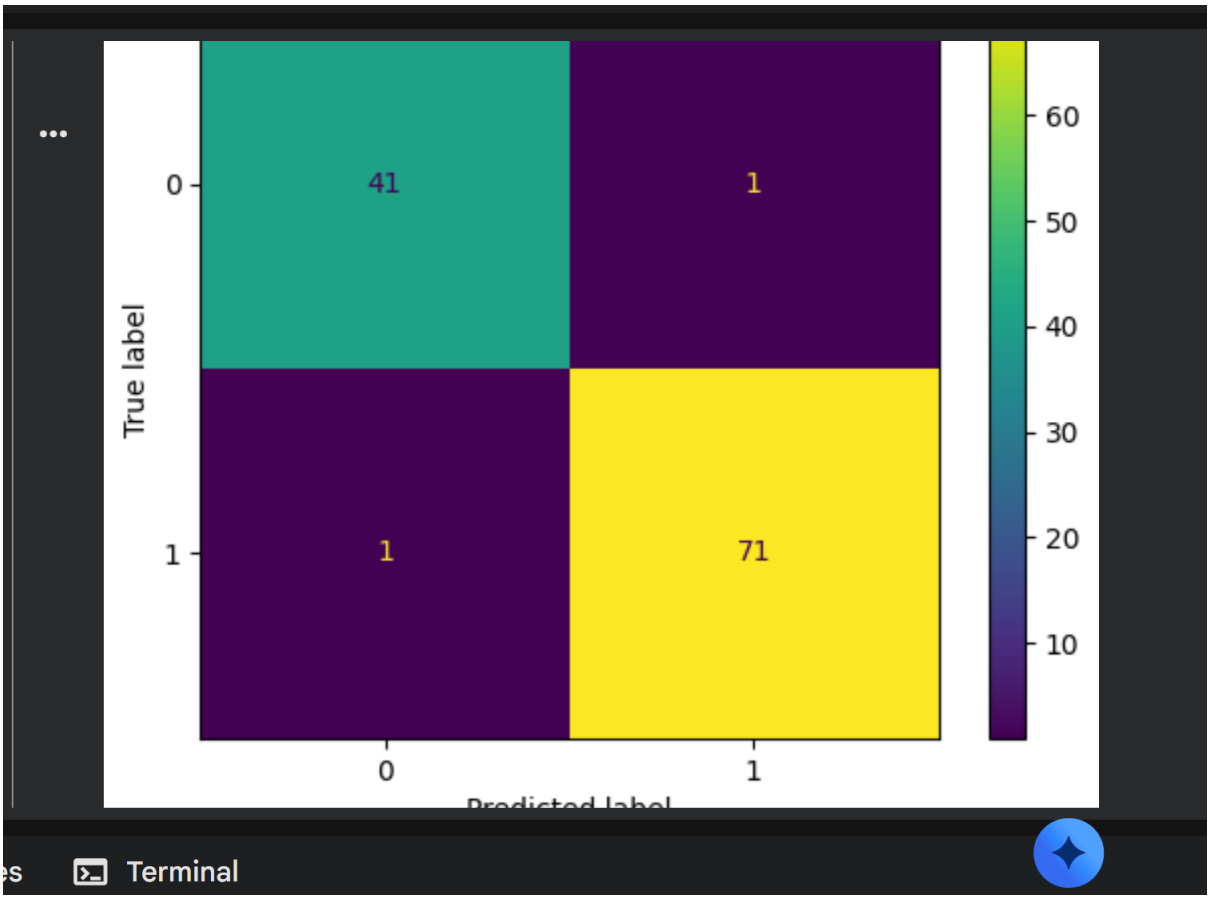
```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print(accuracy, precision, recall, f1)
```

```
0.9824561403508771 0.9861111111111112 0.9861111111111112 0.9861111111111112
```

[7] ✓ 0s

```
from sklearn.metrics import ConfusionMatrixDisplay
import matplotlib.pyplot as plt
ConfusionMatrixDisplay.from_estimator(grid, X_test, y_test)
plt.show()
```





Terminal